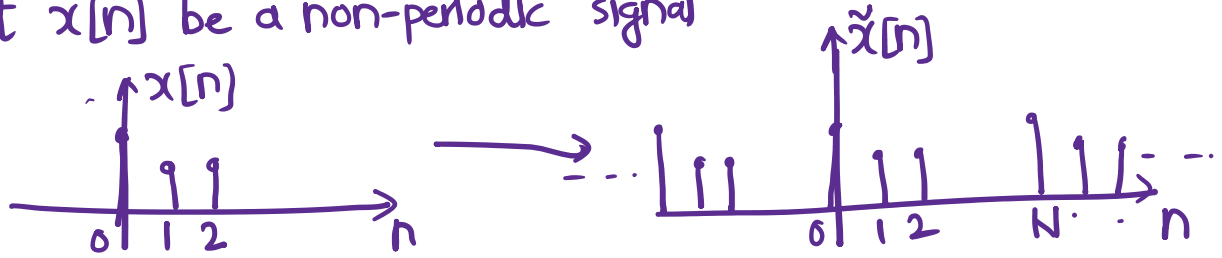
$x[n]$, $x(t)$

Discrete Time Fourier Transform (DTFT)

Let $x[n]$ be a non-periodic signal



$$\tilde{x}[n] = \sum_{k=-\infty}^{\infty} x[n-kN] \quad , \text{ N reasonably large } \Big\} \text{ Periodic with period } N$$

$$\therefore \tilde{x}[n] = \sum_{k=0}^{N-1} a_k e^{jk\omega_0 n} \quad , \quad a_k = \frac{1}{N}\sum_{n=0}^{N-1} \tilde{x}[n] e^{-jk\omega_0 n}$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}$$

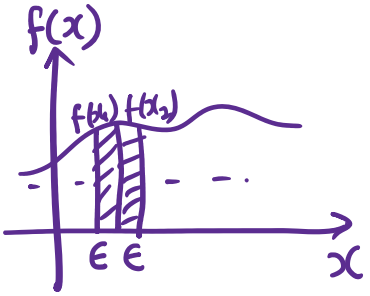
$$a_k = \frac{1}{N}\sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n}$$

Let $X(\Omega) \triangleq \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \Rightarrow a_k = \frac{1}{N} X(k\omega_0)$

$$\therefore \tilde{x}[n] = \sum_{k=0}^{N-1} \frac{1}{N} X(k\omega_0) e^{jk\omega_0 n}$$

$$= \frac{1}{2\pi}\sum_{k=0}^{N-1} X(k\omega_0) e^{jk\omega_0 n} \cdot \frac{2\pi}{N} \quad , \omega_0 = \frac{2\pi}{N}$$

$$\tilde{x}[n] = \frac{1}{2\pi}\sum_{k=0}^{N-1} X\left(k\frac{2\pi}{N}\right) e^{jk\frac{2\pi}{N}n} \cdot \frac{2\pi}{N}$$



Now let $N \to \infty \Rightarrow \frac{2\pi}{N} \to 0$ & $\tilde{x}[n] = x[n]$

$$x[n] = \frac{1}{2\pi}\int_0^{2\pi} X(\Omega) e^{j\Omega n} d\Omega$$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \longrightarrow \text{DTFT Analysis}$$

$$x[n] = \frac{1}{\ldots}\int^{2\pi} X(\Omega) e^{j\Omega n} d\Omega \longrightarrow \text{DTFT Synthesis}$$

$$x[n] = \frac{1}{2\pi}\int_{0}^{2\pi} X(\Omega)\, e^{j\Omega n}\, d\Omega \longrightarrow \text{DTFT Synthesis}$$

<u>Ex.1:</u> $x[n] = a^n u[n]$

$$X(\Omega) = \sum_{n=0}^{\infty} a^n e^{-j\Omega n} = \sum_{n=0}^{\infty} \left(a e^{-j\Omega}\right)^n = \frac{1}{1 - a e^{-j\Omega}}, \quad |a| < 1$$

2) $x[n] = \delta[n]$

$$\Rightarrow X(\Omega) = \sum_{n=0}^{\infty} \delta[n]\, e^{-j\Omega n} = \sum_{n=0}^{\infty} \delta[n] = 1$$

Properties of DTFT:

1) DTFT is linear $\quad x[n] \longrightarrow \boxed{\text{DTFT}} \longrightarrow X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega n}$

2) $X(\Omega)$ is periodic with period 2π.

Pf: $$X(\Omega + 2\pi) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j(\Omega + 2\pi)n} = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega n} \cdot \underbrace{e^{-j2\pi n}}_{1}$$
$$= X(\Omega)$$

3) $x[n]$ is real $\Rightarrow X(-\Omega) = X^*(\Omega)$

Pf: $$X^*(\Omega) = \sum_{n=-\infty}^{\infty} x^*[n]\, e^{j\Omega n} = \sum_{n=-\infty}^{\infty} x[n]\, e^{j\Omega n} = X(-\Omega)$$

$\Rightarrow |X(\Omega)|$ is symmetric

& $\angle X(\Omega)$ is antisymmetric

4) $$x[-n] \longleftrightarrow \sum_{n=-\infty}^{\infty} x[-n]\, e^{-j\Omega n} = \sum_{n'=-\infty}^{\infty} x[n']\, e^{j\Omega n'} = X(-\Omega)$$

5) $x[n]$ is real & even $\Rightarrow X(\Omega)$ is real & even

Pf: $x[n]$ is real $\Rightarrow X(-\Omega) = X^*(\Omega)$

$x[n]$ is even $\Rightarrow x[-n] = x[n] \Rightarrow \underbrace{X(-\Omega) = X(\Omega)}_{\text{even}}$ $\Big\}$ $\underbrace{X(\Omega) = X^*(\Omega)}_{X(\Omega) \text{ is real}}$

$\therefore X(\Omega)$ is real & even.

6) $x[n - n_0] \longrightarrow X(\Omega)\, e^{-j\Omega n_0}$? Time shifting

$\therefore X(\Omega)$ is real & even.

6) $x[n-n_0] \longleftrightarrow X(\Omega)e^{-j\Omega n_0}$ } Time shifting

Pf: $\sum_{n=-\infty}^{\infty} x[n-n_0]e^{-j\Omega n} = \sum_{n'=-\infty}^{\infty} x[n']e^{-j\Omega(n'+n_0)}$, $n-n_0=n'$

$$= \sum_{n'=-\infty}^{\infty} x[n']e^{-j\Omega n'} \cdot e^{-j\Omega n_0}$$

$$= e^{-j\Omega n_0} X(\Omega)$$

7) $e^{j\Omega_0 n}x[n] \longleftrightarrow X(\Omega-\Omega_0)$

Pf: $\sum_{n=-\infty}^{\infty} e^{j\Omega_0 n}x[n]e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x[n]e^{-j(\Omega-\Omega_0)n} = X(\Omega-\Omega_0)$

8) $\overbrace{x[n]*h[n]}^{y[n]} \longleftrightarrow X(\Omega)H(\Omega)=Y(\Omega)$

$$\sum_{k=-\infty}^{\infty} x[k]\underbrace{h[n-k]}_{\frac{1}{2\pi}\int_0^{2\pi} H(\Omega)e^{-j\Omega k}\cdot e^{j\Omega n}d\Omega} = \sum_{k=-\infty}^{\infty} x[k]\frac{1}{2\pi}\int H(\Omega)e^{-j\Omega k}e^{j\Omega n}d\Omega$$

$$\frac{1}{2\pi}\int_{\langle 2\pi\rangle} H(\Omega)e^{j\Omega n}\underbrace{\sum_{k=-\infty}^{\infty} x[k]e^{-j\Omega k}}_{X(\Omega)}\cdot d\Omega$$

$$y[n] = \frac{1}{2\pi}\int_{\langle 2\pi\rangle} H(\Omega)X(\Omega)e^{j\Omega n}d\Omega$$

$Y(\Omega)=H(\Omega)X(\Omega)$

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9) $nx[n] \longleftrightarrow j\frac{dX(\Omega)}{d\Omega}$

Pf: $X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$

$$\frac{dX(\Omega)}{d\Omega} = \sum_{n=-\infty}^{\infty} x[n](-jn)e^{-j\Omega n} = -j\sum_{n=-\infty}^{\infty} nx[n]e^{-j\Omega n}$$

$$= -j\cdot \overbrace{\text{DTFT}\{nx[n]\}}$$

$$\Rightarrow \text{DTFT}\{nx[n]\} = j\frac{dX(\Omega)}{d\Omega}$$

$$\Rightarrow \text{DTFT}\{n x[n]\} = j\frac{dX(\Omega)}{d\Omega}$$

$$\therefore \; n x[n] \longleftrightarrow j\frac{dX(\Omega)}{d\Omega}$$

10) $$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{\langle 2\pi \rangle} |X(\Omega)|^2 \, d\Omega \longrightarrow \text{Parseval's rel}^n$$

$$\sum x[n]\, x^*[n] = \sum x[n] \frac{1}{2\pi}\int_{\langle 2\pi \rangle} X^*(\Omega) e^{-j\Omega n} \, d\Omega$$

$$= \frac{1}{2\pi}\int_{\langle 2\pi \rangle} X^*(\Omega) \underbrace{\sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}}_{X(\Omega)} \, d\Omega$$

$$= \frac{1}{2\pi}\int_{\langle 2\pi \rangle} |X(\Omega)|^2 \, d\Omega$$